

Math Camp: Day 1

Yale University

August 4, 2014

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 - Number of ethnic groups in a country for all countries in the world
 - Highest threshold of per-capita income of a country, in which democracy has collapsed

Sets

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- Sets can numbers as elements, but they can also contain other types of elements: strategies two competing firms might use.

Common Sets:

- $\mathbb{N}$ Natural Numbers
- $\mathbb{Z}$ Integers
- $\mathbb{Q}$ Rational Numbers
- $\mathbb{R}$ Real Numbers
- $\mathbb{C}$ Complex Numbers
- Subscript: $\mathbb{Z}_+$ Positive (Negative) Values of a Set
- Superscript: $\mathbb{Z}^d$ Dimensionality (number of dimensions)

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- The **universal set** contains all elements
- In an **ordered set** the order matters, so sets with the same elements that are listed in different orders are distinct. Sets can also be **unordered sets**.

Operators

Operators: operate of sets and elements of sets

- What are some examples of operators?

Set Operators

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- The **difference** between two sets A and B , denoted $A \setminus B$ is the set containing all of the elements of A that are not in B .
 - If $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$ what is $A \setminus B$?

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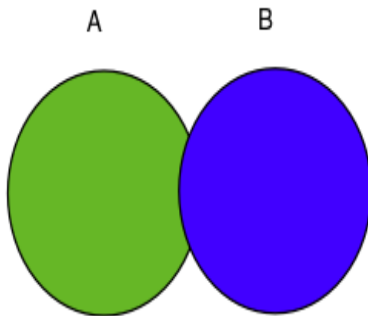
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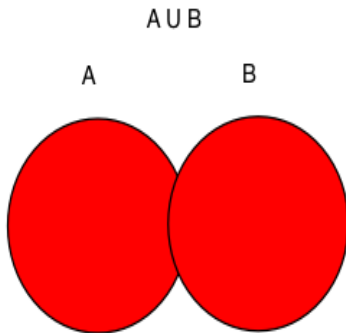
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 - Consider the sets $A = \{\text{Civil War, Peace}\}$ and $B = \{\text{Democracy, Non-Democracy}\}$. What is the Cartesian product of these sets?

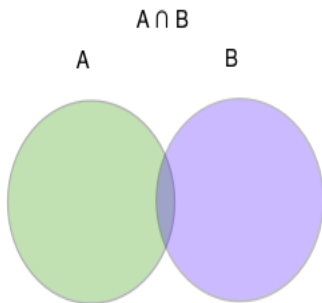
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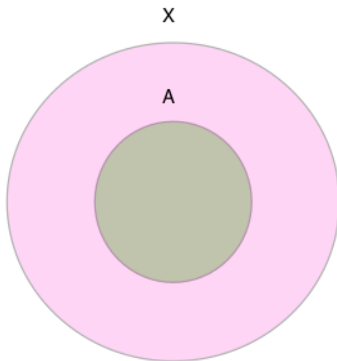
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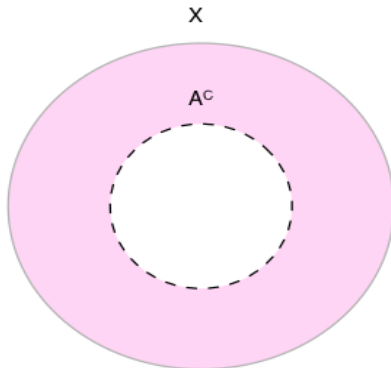
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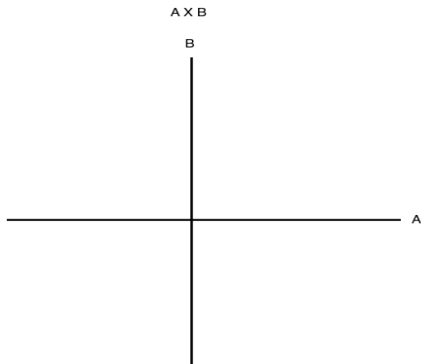
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Levels of Measurement and Relations

What is a a relation? What is an example of a binary relation

Level of Measurement	Mathematical Relation	Meaningful Distance	Zero is Meaningful
Nominal			
Ordinal	x		
Interval	x	x	
Ratio	x	x	x

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- Factorial notation denotes the product of a specific sequence of numbers: $n! = n(n-1)(n-2)\dots(2)(1)$

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- Set of Integers: $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$
- Set of Strictly Positive Integers: $\mathbb{Z}^{++} = \mathbb{N} = \{1, 2, 3, \dots\}$
- Set of Rational Numbers: $\mathbb{Q} = \{x \mid x = \frac{p}{q}, p, q \in \mathbb{Z}, q \neq 0\}$

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- *QED* Latin for “quod erat demonstrandum” or “which was to be proven”.

Proofs

We seek to prove that $A \implies B$. We will discuss five strategies to prove statements:

- Direct Proofs
 - Deductive Reasoning
 - Contrapositive
- Indirect Proofs
 - Contradiction
 - Induction

Deductive Reasoning

Deductive Reasoning relies on a series of accepted axioms or theorems such that

$A_0 \implies A_1 \implies A_2 \implies \dots \implies A_{n-1} \implies A_n$, where $A = A_0$ and $B = A_n$. The crux of this strategy is to find the intermediate steps.

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Proof: A number q is odd if $\exists$ an integer m such that $q = 2m + 1$.
Let $m = 1$. Then $2m + 1 = 3$ Therefore, 3 is an odd number.

Contrapositive

For a contrapositive proof we simply use the fact that $A \implies B$ is equivalent to $\neg B \implies \neg A$.

Example: Let x and y be two positive numbers. Prove that if $xy > 9$, then $x > 3$ or $y > 3$.

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- $\neg B$: Suppose that both $x \leq 3$ and $y \leq 3$.
- $\neg B \implies \neg A$: If $x \leq 3$ and $y \leq 3$, then $xy \leq 9$

Contradiction

We can prove $A \implies B$, by assuming that $A \nRightarrow B$, and finding a contradiction. That is assume that A is true but B is not true. If this yields a contradiction we conclude that: B must be true or A is false. But we have assumed that A is true and so it must be the case that B is true.

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- If $q^2 = 2m^2$, then q is an even number, which is a contradiction.

Induction

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Why does this work? Does the sequence need to be finite?

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- **If $P(k)$, then $P(k+1)$ is true:**
 - Now let $n = k$, and assume that $\sum_{j=1}^k j = \frac{1}{2}k(k+1)$.

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■ **If $P(k)$, then $P(k+1)$ is true:**

□ Now let $n = k$, and assume that $\sum_{j=1}^k j = \frac{1}{2}k(k+1)$.

□ Add $k+1$ to both sides: $\sum_{j=1}^{k+1} j = \frac{1}{2}k(k+1) + k+1$

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Basic Properties of Arithmetic

- Associative: $(a + b) + c = a + (b + c)$
- Commutative: $a + b = b + a$
- Distributive: $a(b + c) = ab + ac$
- Identity: $x + 0 = x$ and $x * 1 = x$
- Inverse: $(-x) + x = 0$ and $x^{-1} * x = 1$ (Not a property of Integers)
- PEMDAS: Please Excuse My Dear Aunt Sally

Ratio, Proportion, and Percentages

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- **Percentage Change:** $\frac{x_{t+1} - x_t}{x_t}$

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- $x = -1$ or $x = -9$

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- Factor the RHS of the equation to see if yields an $(x + \dots)^2$ or $(x - \dots)^2$
- Here we get $(x + 4)^2 = 17$
- Take the Square Root of both sides and simplify to get:
$$x = -4 + \sqrt{17} \text{ or } x = -4 - \sqrt{17}$$

General Solution for Solving Quadratic Equations

The general form of the quadratic equation is $ax^2 + bx + c = 0$.
The general solution to solve for x is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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Functions

A function takes an object (usually a number or an array of numbers) and assigns exactly one other object (usually another number or another array of numbers) to it. For example a utility function transforms consumption into utility: $U(p, b) = 2p + 3b$.

Definition

A *function* f from a set A to a set B is a rule that assigns to each element of A one and only one object in B . It is denoted $f : A \rightarrow B$. The set A is called the *domain* of f and the set B is called the *target space* or *codomain*, and $f(A)$ is called the *range* of f .

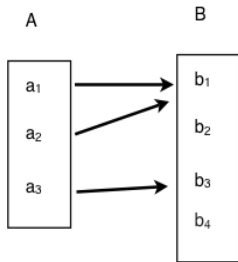
Functions can be **implicitly** defined $y = f(x)$ or **explicitly** defined $f(x) = x^2$.

Function Vs Correspondence

- A function assigns *exactly one* element of set B to an element of set A .
- If the rule assigns more than one element of set B to an element of set A , then the rule is called a correspondence.

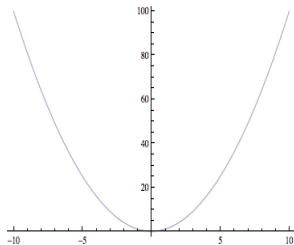
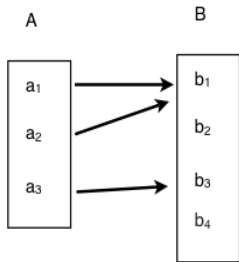
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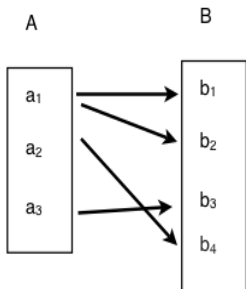
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Graphs

Often we can use graphs to depict Functions and Correspondences. If the relation provides a mapping from the set of real numbers to the set of real numbers $\mathbb{R} \rightarrow \mathbb{R}$, then we can use a Cartesian or xy graph. For example we can graph the function $f(x) = x^2$ or the correspondence $x^2 + y^2 = 1$

Function Composition

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 - For example, $f(x) = 2x$, $g(x) = x^2$, and $h(x) = x + 2$. Then $g \circ h(x) = (x + 2)^2$ and $f \circ (g \circ h(x)) = 2(x + 2)^2$.
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 - But it is also the case that $f \circ g(x) = 2x^2$ and $(f \circ g) \circ h(x) = 2(x + 2)^2$
- Composition of functions are not always commutative. It is not always the case that $f(g(x)) = g(f(x))$. For example, consider the case when $f(x) = 3x$ and $g(x) = x^2$.

Injective, Surjective, and Bijective

- The function f is called *injective* or *one-to-one* if $\forall b \in f(A), \exists$ only one $x \in A$ with $b = f(x)$. This means that every output of the function has at most one input. Or for any $x, y \in A$, if $f(x) = f(y)$, then $x = y$.

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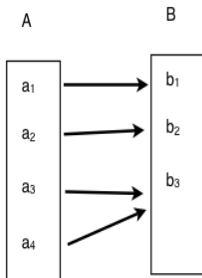
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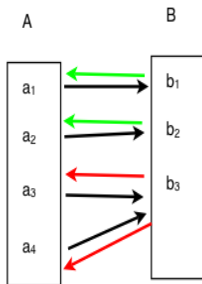
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- If a function is onto and one-to-one we can define the inverse function f^{-1}

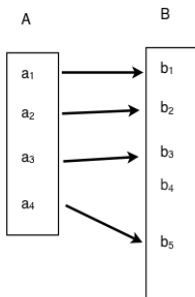
Onto and One-to-One, Examples



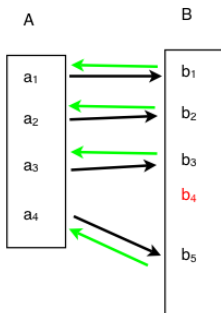
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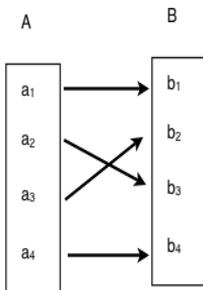
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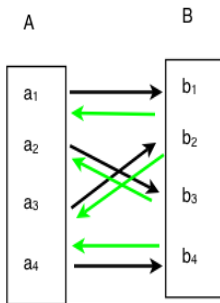
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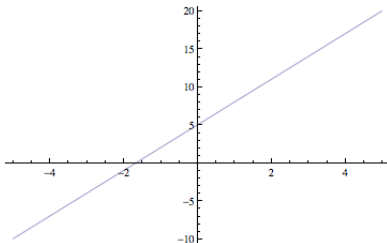
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- **Boundedness** The function f is bounded if $\exists K \in \mathbb{R}$ such that $|f(x)| < K \ \forall x \in X$.

Functions in more than one Variable

Functions can provide a mapping from $A_1 \times A_2 \times \dots \times A_n \rightarrow B$. For example, $f(x, y) = x^2 + 3y$ or $f(x, y, z) = xy + 10z$. If the domain is two dimensions then the function can be graphed on a 3 dimensional plot.

Linear and Affine Functions

An affine function or a takes the shape of $y = a + bx$, where a and b are constants. This function is useful when the effects of x on y are constant. For example, $y = 5 + 3x$:



Technically this is not a linear function, which does not permit the constant, a . See the details on page 55-56.

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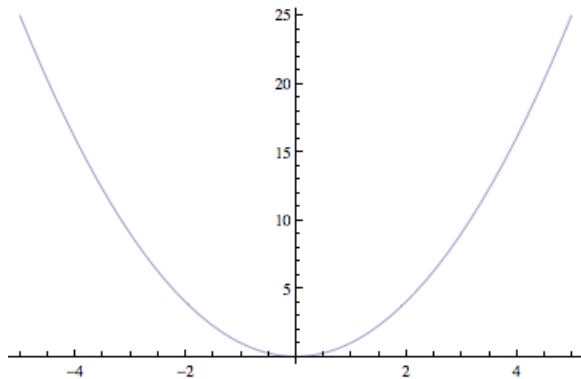
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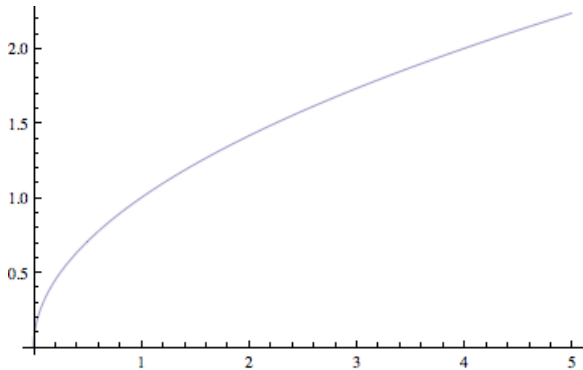
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- $\frac{x^n}{x^m} = x^{n-m}$, since $x^{-m} = \frac{1}{x^m}$

Plotting $f(x) = x^2$



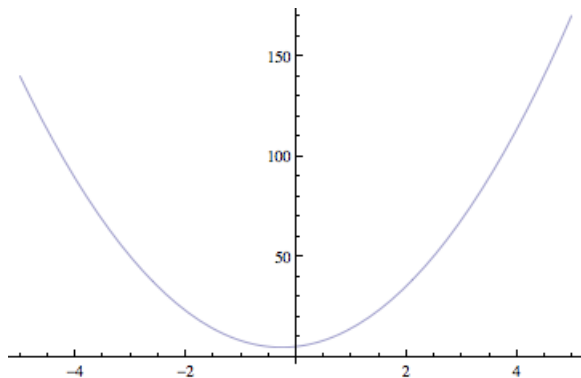
Plotting $f(x) = x^{\frac{1}{2}}$



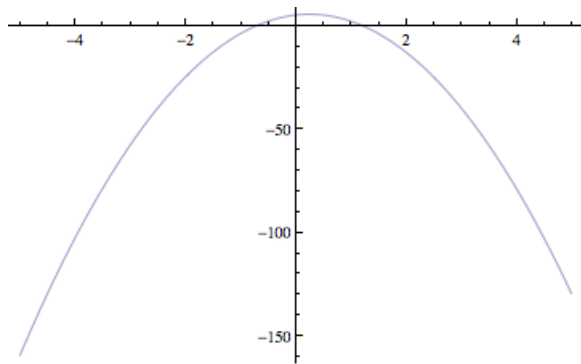
Quadratic Functions

If y is a **quadratic function** of x then $y = \alpha + \beta_1 x + \beta_2 x^2$. This is a non-linear function that describes a parabola. These functions allow one to depict a “U” relationship between variables. For example, protests might be less likely to arise in non-repressive settings and in highly repressive settings.

Plotting $y = 5 + 3x + 6x^2$



Plotting $y = 5 + 3x - 6x^2$

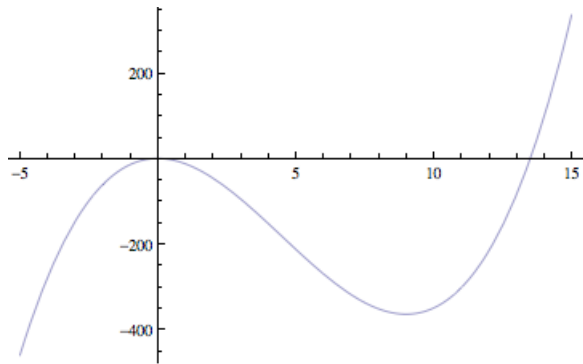


Higher-Order Polynomials

Polynomial Functions have the general form

$y = \alpha + \beta_1 x + \beta_2 x^2 + \dots + \beta_n x^n$, where $n < \infty$. These functions are not extremely common in applied political science work but are useful for later material. They allow one to depict relationships between variables when there are two or more thresholds over which the relationship between variables changes.

Plotting $y = x^3 - \frac{27}{2}x^2$



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- If $\log_a x = b$, then $a^{\log_a x} = a^b$

Log and ln

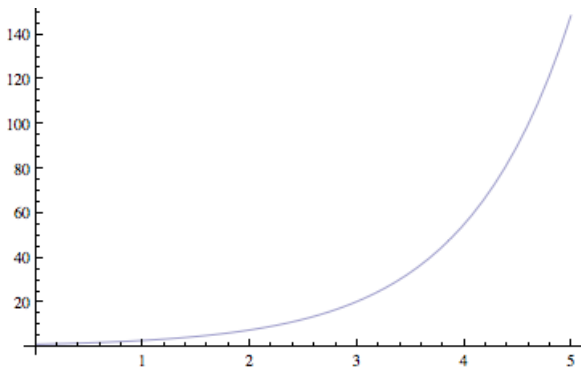
The two most common bases for a logarithm are 10 and $e \approx 2.7183$. The function $f(x) = e^x$ has many important properties that we will use later. It is also used, among other reasons, to calculate compound interest and populations of animals. Often the log with a base of e is written as $\ln$, but it is also written as $\log$. For $\ln$, we have

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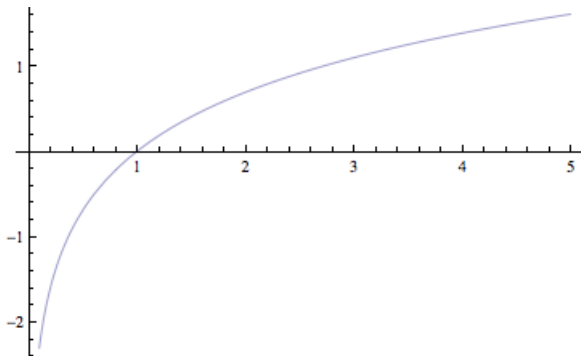
■ $e^{\ln x} = x$ and $\ln(e^x) = x$

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This function is useful for depicting a relationship between two variables that increases at an increasing rate: Examples?.

Plotting $f(x) = \ln(x)$



This function is useful for depicting a relationship between two variables that increases at a decreasing rate: Examples?

Properties of Logarithms

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- Properties can be useful for transforming equations into simpler ones: Cobb Douglas Utility Function can be written as $U(p, b) = p^x b^{(1-x)}$ or the natural log of the utility function $u(p, b) = x \ln(p) + (1 - x) \ln(b)$

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$$f(x) = \begin{cases} x + 5 & : x \leq 6 \\ x^2 & : x > 6 \end{cases}$$

- If is no chance of a democracy collapsing if GDP per capita exceeds \$6055, but that the likelihood increases as GDP per capita decreases below this level. Then we might write a function that defines the likelihood of collapse as a piecewise function.

Concavity and Convexity

- **Concavity:** The function f is *concave* if $\forall \alpha \in [0, 1]$ and $x_1, x_2 \in \mathbb{R}$, $f(\alpha x_1 + (1 - \alpha)x_2) \geq \alpha f(x_1) + (1 - \alpha)f(x_2)$
- **Convexity:** The function f is *convex* if $\forall \alpha \in [0, 1]$ and $x_1, x_2 \in \mathbb{R}$, $f(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha f(x_1) + (1 - \alpha)f(x_2)$
- These are different concepts than a convex set

Preference Relations

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Property 1 of Rational Preferences: Completeness (or Comparability)

- Preference relations are complete if for any two alternatives (say, Z and L) it is the case that ZP_iL , LP_iZ , or ZI_iL .
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Maximum Principle

- With completeness and transitivity an individual can order alternatives from the lowest to the highest alternative. A rational actor will select the highest alternative.

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- Alternatively we could model the utility function as $u_i(x)$ and give it properties such as $u_i(x)$ is concave and monotonically increasing.

Best Response Correspondences

Consider two states, s_1 and s_2 that can choose to invest in guns G or invest in domestic spending B . Say that s_1 is a hostile state that always prefers to invest in guns. Say that s_2 prefers to invest in guns if s_1 invests in guns, but prefers to invest in domestic spending if s_1 invests in domestic spending.

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 - Now we must write a **best response correspondence**, since $B_{s_1}(B) = \{G, B\}$